

Flow Dynamics of Lunar Regolith For In-Situ Utilization and Transportation

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Lunar regolith, the loose, heterogeneous material covering the Moon's surface, presents both challenges and opportunities for extraterrestrial construction. This research aims to study the fundamental mechanics of Lunar regolith flow using ANSYS Rocky. Renowned for simulating bulk materials with intricate particle shapes and diverse sizes, ANSYS Rocky models particle-particle interactions accurately. The focus lies in analyzing regolith simulant behavior through pipe-like structures under varying physical environments such as different gravitational forces. Simulations will be validated with physical experiments, demonstrating Rocky's efficacy in simulating particle motion on the Moon. This marks an initial step toward achieving space-based in-situ utilization and transportation.

I. Introduction

THE exploration and utilization of extraterrestrial resources have emerged as paramount objectives in contemporary space endeavors, particularly in the context of establishing sustainable human habitats beyond Earth. Considering NASA's recent emphasis on returning humans to the Moon through the Artemis program, specifically targeting the Lunar South Pole, focus is geared towards establishing a sustainable presence on and around the Moon - a lasting and productive presence enabled by reusable systems [1]. The Lunar South Poles' terrain, scattered with craters, which create permanently shadowed areas that remain at very low temperatures, capturing resources like water ice, oxygen, and hydrogen [2], has presented itself as an area of interest with an array of challenges. One recent highlight of the landscape's challenges occurred during the final descent of Intuitive Machines' IM-1 Lunar Lander. On February 22nd, 2024 the IM-1 Lander, Odysseus, experienced an anomaly upon touchdown, tipping over and coming to rest on its side due to the powdery and unstable properties of the Lunar regolith. These unfavorable conditions This was a result of the micrometeorite impacts that lead to insufficient particle sorting [3]. The fine, loose material failed to provide adequate support during touchdown, a fatal consequence for the Lander.

In response to the growing interest of the Moon's Southern regions, manufacturing landing pads is necessary to conduct safe landings and launches from the Lunar surface. Dust ejecta from a rocket plume can, for instance, erode nearby coated surfaces and get into mechanical assemblies near the landing sites [4] or cause the landing area to be non-reuseable. Therefore, a strong and resilient platform enhances the stability of landers, enabling continuous and reliable usage for ongoing Lunar exploration and operations. The magnitude of construction material required to create hardened surfaces is substantial; therefore, it is imperative to identify an abundant resource already present on the Moon. Lunar regolith, readily available on the Moon's surface, presents an ideal candidate for this purpose. By employing in-situ resource utilization (ISRU) techniques, specifically in situ manufacturing with regolith, it is possible to create these structures efficiently and autonomously. Beyond landing pads, new manufacturing technology may be applied to build other essential structures through ISRU, such as roads and radiation shelters, significantly reducing cost, mass, and risk of sustained human activities beyond low-Earth orbit (LEO) [5].

To realize the potential of Lunar regolith for such applications, extensive research has already been conducted to explore its use in manufacturing processes. Various studies have focused on leveraging in-situ resource utilization (ISRU) methods to transform regolith into durable materials suitable for building essential infrastructure on the Moon. For instance, there is a patented technology called D_Shape that has already demonstrated the ability to build large-scale construction artifacts. This technology is being explored for potential use in lunar applications, where lunar regolith can be combined with a special "ink" designed to agglomerate the particles, effectively binding them together [6]. On the other hand, rather than relying on 3D printers and binder materials that need to be transported from Earth, other research has focused on manufacturing structures on the Moon using only lunar regolith, sunlight, and the vacuum environment. This approach explores the possibility of using the natural resources present on the Moon, combined with

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minimal communication with Earth to develop a telepresence system with an unmanned robot. This telepresence would enable remote operation and initiate a flow of construction, allowing for autonomous building processes on the lunar surface without the need for extensive Earth-based intervention [7]. While these various approaches offer potential solutions for utilizing lunar regolith in construction, there are still decisions to be made determining the appropriate machinery for regolith manipulation, selecting suitable binder materials, and ensuring access to the necessary resources for the construction process. Despite the different strategies being explored, one fundamental requirement remains consistent: comprehensive testing must first be conducted in an accessible Earth-based environment. This testing is essential to evaluate the feasibility of these methods under controlled conditions before they are deployed in the harsh lunar environment. Simulations must account for the Moon's unique challenges, including its vacuum, extreme temperature fluctuations, and abrasive regolith, to ensure that these construction techniques are practical and reliable for sustained lunar operations. Additionally, experimenting with lunar regolith is challenging due to its scarcity.

Due to Lunar regolith samples available on Earth are extremely limited and primarily reserved for non-destructive research, simulated experiments help demonstrate the importance of allocating some of this material for destructive testing, such as sintering or melting, to advance in-situ resource utilization techniques for Lunar exploration. ANSYS Rocky, a powerful particle simulation tool, is being used in this research to model the behavior of Lunar regolith under various conditions, enabling researchers to optimize processes and minimize resource waste before conducting physical tests. This validation serves as critical evidence to justify the use of authentic Lunar regolith for further testing.

The following sections provide an explanation of the theory and methodologies underpinning the current simulations. Section II delves into the particle modeling approach and the properties of the materials simulated in ANSYS Rocky, detailing the characteristics of volcanic ash and the Lunar regolith simulant. Section III outlines the static angle of repose test conducted with these modeled particles and describes the physical experiment performed using Lunar regolith to validate the simulation results. Section IV leverages the static angle of repose findings to highlight the importance of sintering the particles and investigates the behavior of sintered particles under Earth's and Lunar gravity conditions. Lastly, Section V discusses the key results and insights derived from this research proposing potential improvements and outlines the next steps for future studies.

II. Particle Modeling/Properties

To achieve reliable results without the need to waste valuable physical materials, scientifically validated simulation tools play a crucial role. Particle modeling capabilities allow for the exploration of various scenarios for utilizing Lunar regolith. This modeling is accomplished by gathering the particles' exact geometry, ensuring accurate physical representation of their properties. To further refine the particle characterization, sieve analysis was conducted, which involves passing granular particles through a series of sieves with standardized aperture sizes. Cumulative distribution complements this process by showing the cumulative percentage of material passing through each sieve, providing a clear picture of particle size gradation [8]. As a result, cumulative distribution graphs were analyzed and utilized in this study to accurately represent particle size distributions. This approach helps refine target experiments by narrowing the scope to scientifically validated methods. By relying on proven results, it is possible to identify effective techniques while minimizing the risk of inefficiently using or wasting the limited regolith resources available for testing. This streamlines the development process and ensures that critical resources are preserved for the most promising applications.

The primary objective of this investigation is to employ ANSYS Rocky, a discrete element modeling (DEM) particle simulation software [9], renowned for its ability to precisely simulate the flow dynamics of bulk materials featuring intricate particle shapes and diverse size distributions. The software displays particle-particle interactions by modeling every particle's motion, avoiding the need for constraint equations [10]. All particles within the computational domain are simulated using a Lagrangian approach, treating each particle as an individual entity and computing their movement over time based on the forces acting upon them. This method explicitly solves Newton's second law of motion, which governs particle behavior in terms of translation (linear movement) and rotation (spinning motion). The translational dynamics, which describe how the particle's velocity changes under the influence of various forces, are given by:

$$m_p \frac{d\mathbf{v}_p}{dt} = \mathbf{F}_p^C + m_p \mathbf{g} + \mathbf{F}_p^{fp} \quad (1)$$

Where m_p is the particle mass, $\mathbf{v}_p$ is the velocity, $\mathbf{F}_p^C$ represents the contact force, $m_p \mathbf{g}$ is the gravitational force, and $\mathbf{F}_p^{fp}$ accounts for any fluid-particle interaction forces. This method also considers the rotational behavior of particles, governed by Newton's second law for rotational motion. The rotational dynamics describe how the particle's angular velocity changes due to applied torques and are expressed as:

$$\mathbf{I}_p \frac{d\boldsymbol{\omega}_p}{dt} = \mathbf{T}_p \quad (2)$$

Where I_p represents the moment of inertia tensor, ω_p is the angular velocity vector, and T_p denotes the applied torque. [11].

Our goal is to use this advanced software to closely analyze the behavior of regolith simulant particles as they flow through various pipe-like structures in different tests. This paper examines how different physical environments, such as gravity, hopper dimensions, and particle size, affect these flows. Computer simulations will be validated with experiments, providing evidence that ANSYS Rocky is an effective tool for simulating particle motion on the Moon. These smaller particles significantly increase the computational complexity, as they require precise calculations to account for their interactions with other particles and their surroundings. To handle intensive simulations, the Cray CS400 supercomputer is utilized [12]. With its 192 CPU core processors, VEGA is capable of managing the immense computational load, ensuring accurate and efficient simulation of every particle's behavior.

To bridge the gap between simulated and experimental results, volcanic ash, a terrestrial material with properties similar to Lunar regolith, is modeled as an initial benchmark. This approach enables direct comparisons with physical test results, allowing validation of the simulation tools' reliability while offering a more accessible alternative for researchers. By utilizing volcanic ash, it is possible to conduct comparable studies without depending on the scarce Lunar regolith or its simulant, JSC-1A. Following this, the Lunar regolith simulant JSC-1A is used due to its close resemblance to authentic Lunar regolith in composition and behavior. While the simulant is inspired by volcanic ash, it has been specifically engineered to replicate the unique characteristics of Lunar soil, making it a valuable stand-in for the limited supply of authentic Lunar regolith. Figure 1 illustrates a flowchart outlining the configuration settings for particle modeling. The process begins with defining the material properties, including density, Young's modulus, and Poisson's ratio, as depicted on the left. Subsequently, the material properties are assigned to a newly created particle, modeled either as a simple spherical shape or using a computer-aided design (CAD) file to define its geometry, and then configured to accommodate various size distributions utilizing ANSYS Rocky's sieve size modeling.

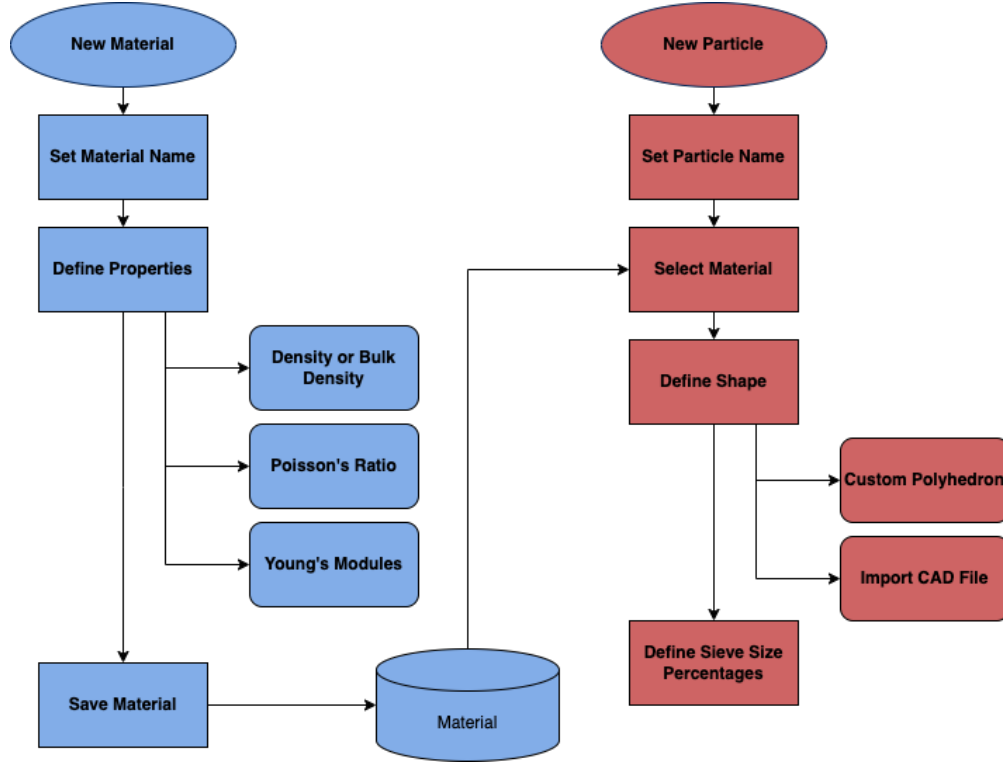


Fig. 1 ANSYS Rocky particle configuration flow

A. Volcanic Ash Properties

Volcanic ash from Mt. St. Helens is chosen as the comparison material for Lunar regolith due to the significant similarities in mineral and chemical compositions. Both volcanic ash and Lunar regolith share a predominance of silicate minerals, which constitute the majority of their mineral content. Silicate minerals, such as oxides and aluminosilicates, are abundant in both materials, making the volcanic ash a suitable candidate for simulating the conditions found on the Moon. [13], [14]. In terms of composition, the mineral profiles of these two materials exhibit significant overlaps, especially in key components such as silicon dioxide (SiO_2), aluminum oxide (Al_2O_3), and iron oxide (FeO), which are critical for understanding the behavior of regolith in various processes, including sintering and melting. The similarity in composition provides an opportunity to explore the characteristics and behavior of Lunar regolith simulants, like JSC-1A, by using volcanic ash as a proxy material for simulation in particle modeling. Table 1 below presents a direct comparison of the mineral compositions of volcanic ash and Lunar regolith. It highlights the percentage of each mineral present in both materials, demonstrating their shared characteristics.

Table 1 Mineral composition comparison of volcanic ash and Lunar regolith

Mineral Compositions		
Mineral	Volcanic Ash	Lunar Regolith
SiO_2	65%	43%
Al_2O_3	16%	18%
FeO	5%	12%
CaO	4%	11.7%
MgO	2%	10.3%

The simulation inputs shown in Figure 2, utilize the exact grain shape for volcanic ash from Mt. St. Helens [15]. Both the volcanic ash and Lunar regolith are fine-grained with particle sizes ranging from very fine dust to small, irregularly shaped particles. Their textures are also similar due to their formation processes, with both containing rough, angular particles from rapid cooling and fracturing, which further supports the use of volcanic ash as a comparable material to lunar regolith. Using this particle makeup, ANSYS Rocky's Sieve Size method was applied using the cumulative particle size distribution data found in Figure 3. The samples of volcanic ash was collected in six different zones, denoted by 'z'. [16]. For this experiment, the volcanic ash in zone 'z3' was chosen due to its larger particle size distribution compared to other zones. Larger particles are more suitable for ANSYS Rocky simulations as they not only enhance the accuracy of flow behavior predictions but also reduce computational time, allowing for quicker simulations without compromising the reliability of the results.

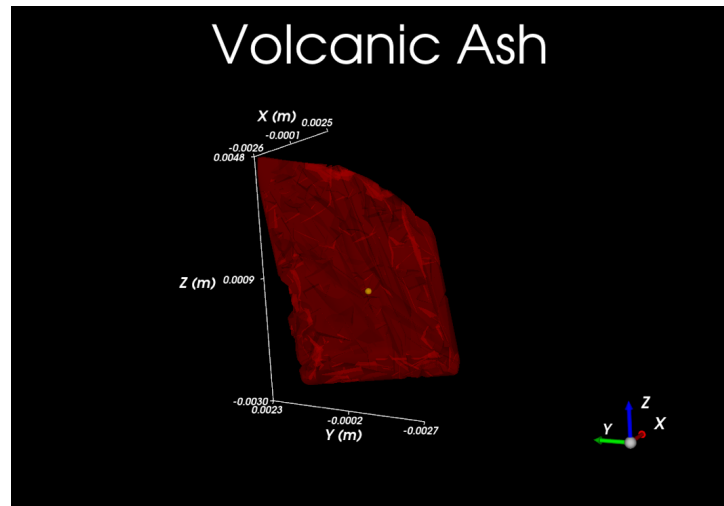


Fig. 2 Volcanic ash particle importated into Rocky

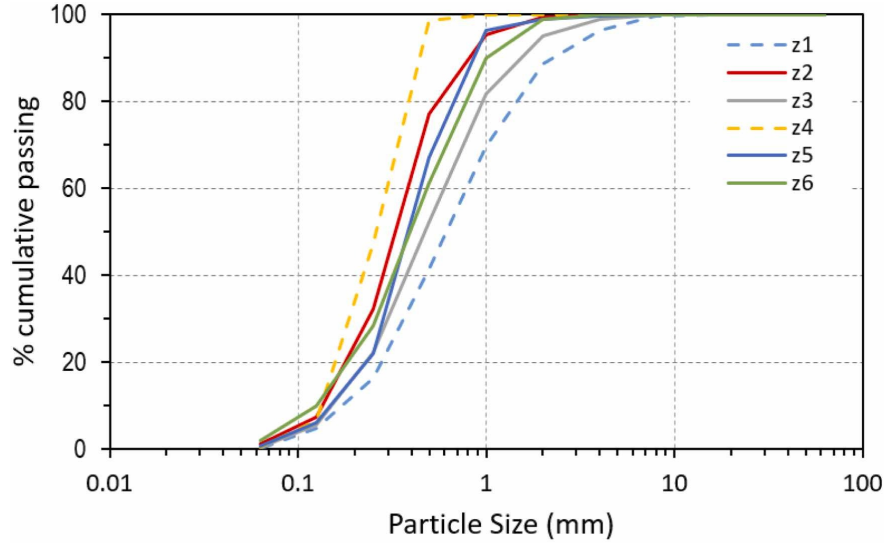


Fig. 3 Particle size of the samples of volcanic ash

B. Lunar Regolith Simulant: JSC-1A Properties

While volcanic ash, such as that from Mt. St. Helens, shares key silicate minerals with lunar regolith and offers an accessible and cost-effective alternative for initial studies, it lacks some of the specific characteristics found in lunar materials. For example, volcanic ash has larger particle sizes and different mineral compositions, which do not accurately replicate the fine-grained and highly angular nature of lunar regolith. JSC-1A, on the other hand, is a Lunar regolith simulant derived from basaltic volcanic ash, designed to closely replicate the composition and characteristics of the Lunar soil. The simulant was specifically engineered to mimic the properties of lunar mare soil, including its chemical composition, mineralogy, particle size distribution, specific gravity, and cohesion [17].

For the simulation inputs shown in Figure 4, exact grain data for the JSC-1A Lunar regolith simulant particles was gathered, using CAD files from approximately 14,000 particles studied via X-ray micro-CT and nano-CT scans [18]. Using this particle makeup, ANSYS Rocky's sieve size method was applied using the cumulative particle size distribution data found in Figure 5 for the JSC-1A Lunar regolith simulant [19].

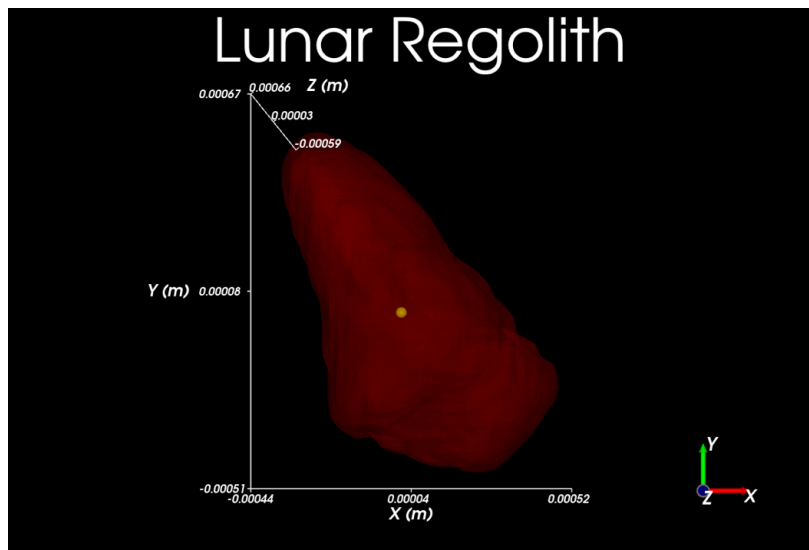


Fig. 4 JSC-1A particle importated into Rocky

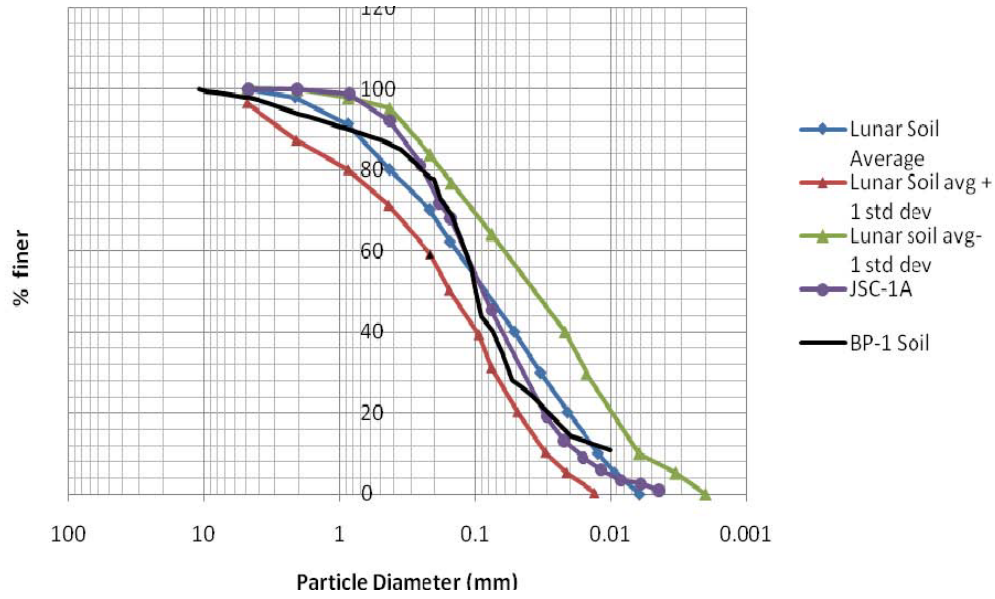


Fig. 5 Particle size distribution of Lunar soil, JSC-1A, and BP-1 soil

III. Static Angle of Repose Test

The angle of repose is the maximum angle at which grains can remain stable without shifting. It can be categorized as static or dynamic based on the grain's state of motion. The static angle of repose (SAOR) can be measured just before and after slope instability, which is useful for understanding the maximum stable slope a granular material can maintain while at rest. The dynamic angle of repose (DAOR) can be measured when grains are moving continuously down an inclined plane, useful for analyzing the behavior of a granular material as it is being poured[20]. This study focuses on the static angle of repose of the Lunar regolith particles, assisting in understanding how the material will settle and compact before the sintering process. It can be measured simply by pouring the regolith to form a conical pile inside a cylinder, then lifting the cylinder up to create a measurable pile [21]. Understanding the static angle of repose and flow behavior of particles is essential for assessing the suitability of scarce materials like Lunar regolith in various applications, including additive manufacturing.

By using ANSYS Rocky to simulate particle behaviors, it is possible to directly evaluate how cohesive properties impact material flow. This tool models particle interactions in detail, allowing for a comprehensive analysis of dynamics such as clogging and inconsistencies in powder feeding. Cohesive particle behavior can restrict flow from a hopper, making it more difficult for particles to exit and increase the risk of clogging. In contrast, particles that are more free-flowing improve hopper flow, as they form a flatter pile when poured and move easily through machinery. Table 2 illustrates the relationship between particle flow-ability and its angle of repose. Particles with lower repose angles exhibit greater flow-ability, while higher angles indicate increased cohesiveness [21]. For this experiment, achieving a free-flowing nature is critical to ensure consistent powder feeding, important in applications such as additive manufacturing.

Table 2 Carr classification of flowability of powder based on repose angle

Description	Repose Angle
Very free-flowing	<30°
Free flowing	30–38°
Fair to passable flow	38–45°
Cohesive	45–55°
Very cohesive (non-flowing)	>55°

The angle of repose test was simulated in ANSYS Rocky using volcanic ash and the Lunar regolith simulant JSC-1A, as modeled in the previous section. However, despite leveraging the supercomputer's advanced computational capabilities, the simulation of large quantities of detailed CAD particles proved infeasible due to the immense computational load. To address this limitation, spherical particles were used as a simplified representation for the analysis, ensuring computational efficiency while maintaining an acceptable level of accuracy for the study's objectives. Following the simulation results, a physical test was conducted to validate the accuracy of the simulated JSC-1A properties. This evaluation assessed the reliability of the ANSYS Rocky simulation tool, ensuring it accurately represents the mechanical and flow behaviors of the simulant. Figure 6 illustrates the flowchart detailing the configuration settings required for modeling the static angle of repose of a particle group. Here, the static angle of repose test will involve using a hollow cylinder with a 1 mm diameter and a height of 3 mm as an essential "wall" component in the simulation setup. A volumetric inlet is then configured, utilizing the hollow cylinder as the containment shape. This inlet is filled entirely with the simulated particles, modeled using a CAD particle assembly. The particle distribution is defined based on sieve size inputs derived from cumulative particle distribution graphs, ensuring accurate representation of the particle properties within the simulation environment. Following this setup, a frame of motion is applied to the hollow cylinder. A translational motion moves the cylinder upward along the y-axis, allowing the particles within to disperse freely. As the particles flow out, they form a pile, which can then be analyzed to measure the angle of repose. This process provides insights into the material's flow characteristics, aiding in the evaluation of particle behavior under simulated conditions. To test for free-flowing behavior, Earth's gravity will be used for comparison, providing a benchmark to validate the reliability of Rocky utilizing the Carr classification in Table 2.

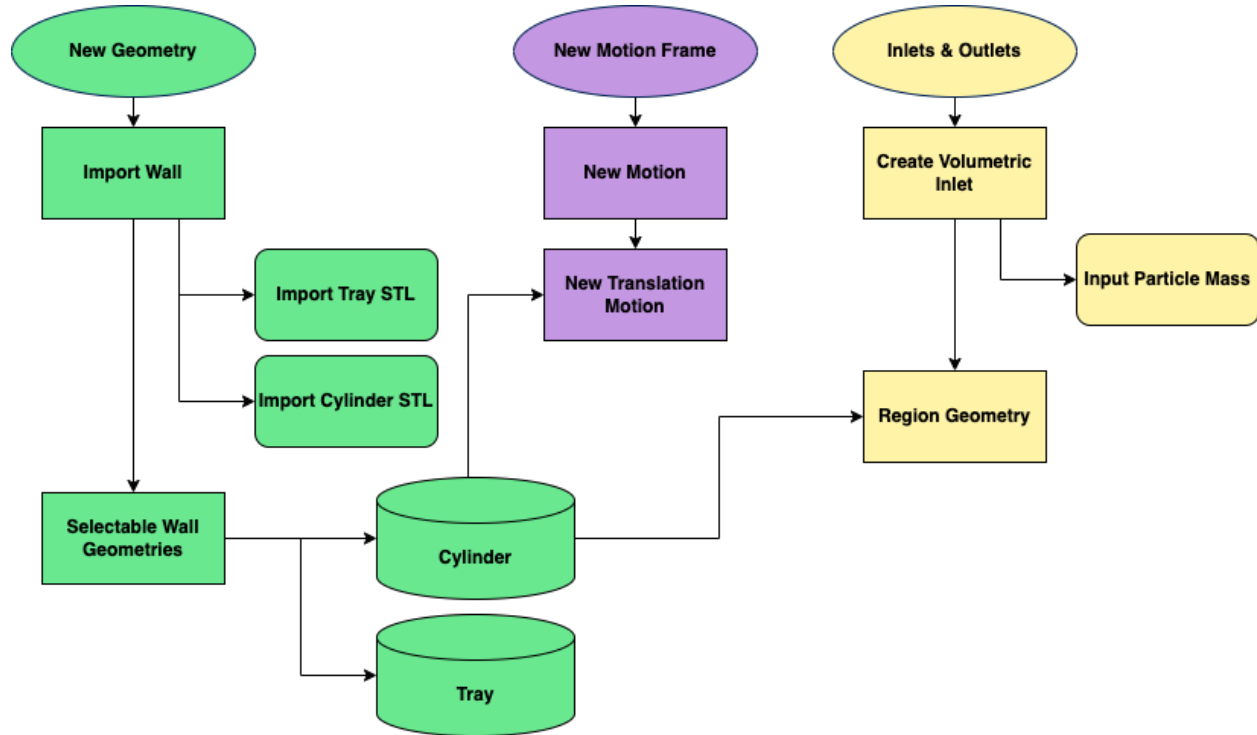


Fig. 6 ANSYS Rocky SAOR configuration flow

A. SAOR Volcanic Ash: Earth's Gravity Simulation

Using the SAOR test case for volcanic ash in Earth's gravity, it is possible to observe the behavior of the material as it escapes the cylinder and accumulates on a downstream surface. This information is extracted from the resulting particle pile by mapping the pile's cross-section within the X-Y cartesian coordinates, as seen in Figure 7. The angle of repose for the particle dispersion is determined using the arc tangent function, giving an angle of repose of about 12 degrees, which when comparing to the Carr classification proves that the volcanic ash is very free-flowing.

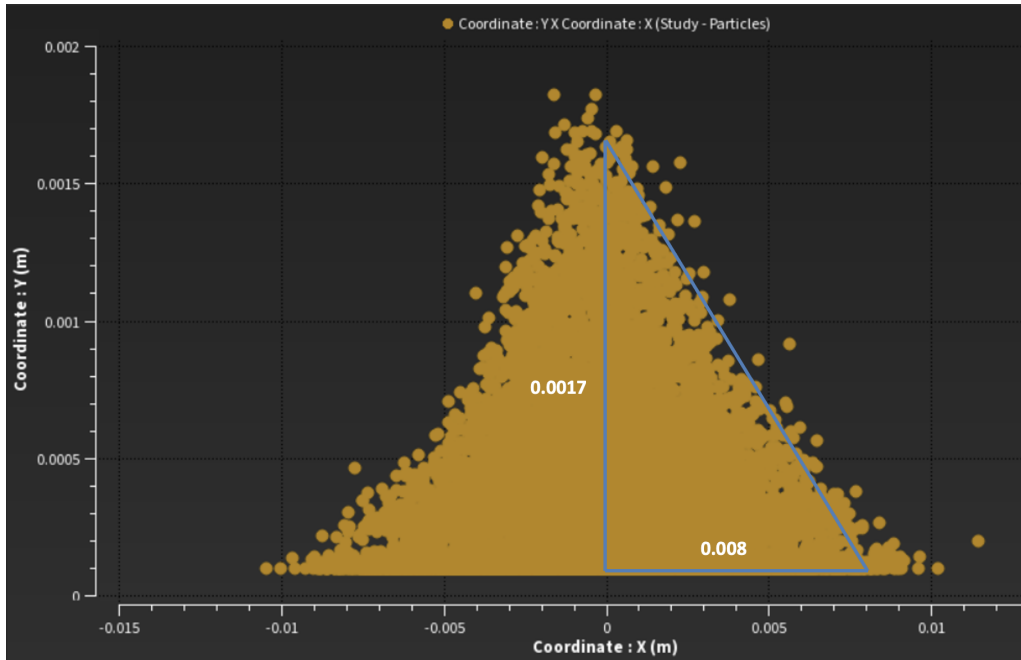


Fig. 7 Static angle of repose particles for volcanic ash

B. SAOR JSC-1A: Earth's gravity simulation

Switching to the simulation of JSC-1A particles, the material behavior is analyzed as it exits the cylinder under the same configurations detailed in the flowchart from Figure 6. The accumulation of particles is examined by mapping the pile's cross-section within the X-Y cartesian coordinates, as shown in Figure 8. Utilizing the arc tangent formula again, the Lunar simulant's angle of repose showed to be about 4 degrees, which when comparing to the Carr classification tells us that the JSC-1A simulant is also very free-flowing, making it ideal for consistent powder feeding in processes like additive manufacturing.

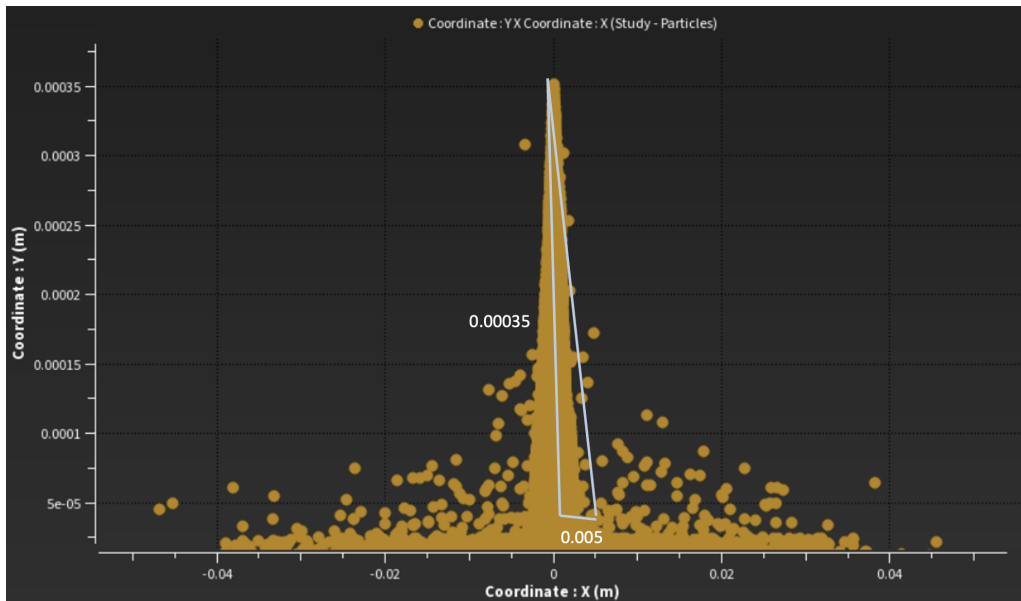


Fig. 8 Static angle of repose particles for JSC-1A

The particle differences observed in the simulation graphs reveal that volcanic ash forms a neater and more uniform pile compared to JSC-1A. This discrepancy arises primarily from the differing material quantities used in the tests. Volcanic ash, with its larger particle size distribution, filled a cylinder with a 3 mm radius, whereas JSC-1A required a smaller 1.5 mm radius cylinder, fully packed. The static angle of repose (SAOR) test is slightly sensitive to sample size, with larger sample volumes generally yielding more reliable results. However, as demonstrated in the physical test in the following section, the differences between the simulated and physical JSC-1A results are minimal, suggesting that the test remains fairly consistent despite variations in the material quantities used.

C. SAOR JSC-1A: Physical Experiment

To compare simulated results, a static angle of repose test was conducted using a physical setup with slight adjustments. The experiment utilized JSC-1A regolith simulant from Johnson Space Center as the test material. The setup featured a slightly larger hollow cylinder, measuring 4 mm in diameter and 4 mm in height. The cylinder was filled to half its height with particles. Mirroring the simulation, the cylinder was lifted vertically in the y-direction, allowing the grains to fall freely and settle. This process facilitated a direct comparison between the physical experiment and the simulated results, enhancing the validation of the modeling approach. Figure 9 displays the particle pile, viewing from an X-Y coordinate plane.

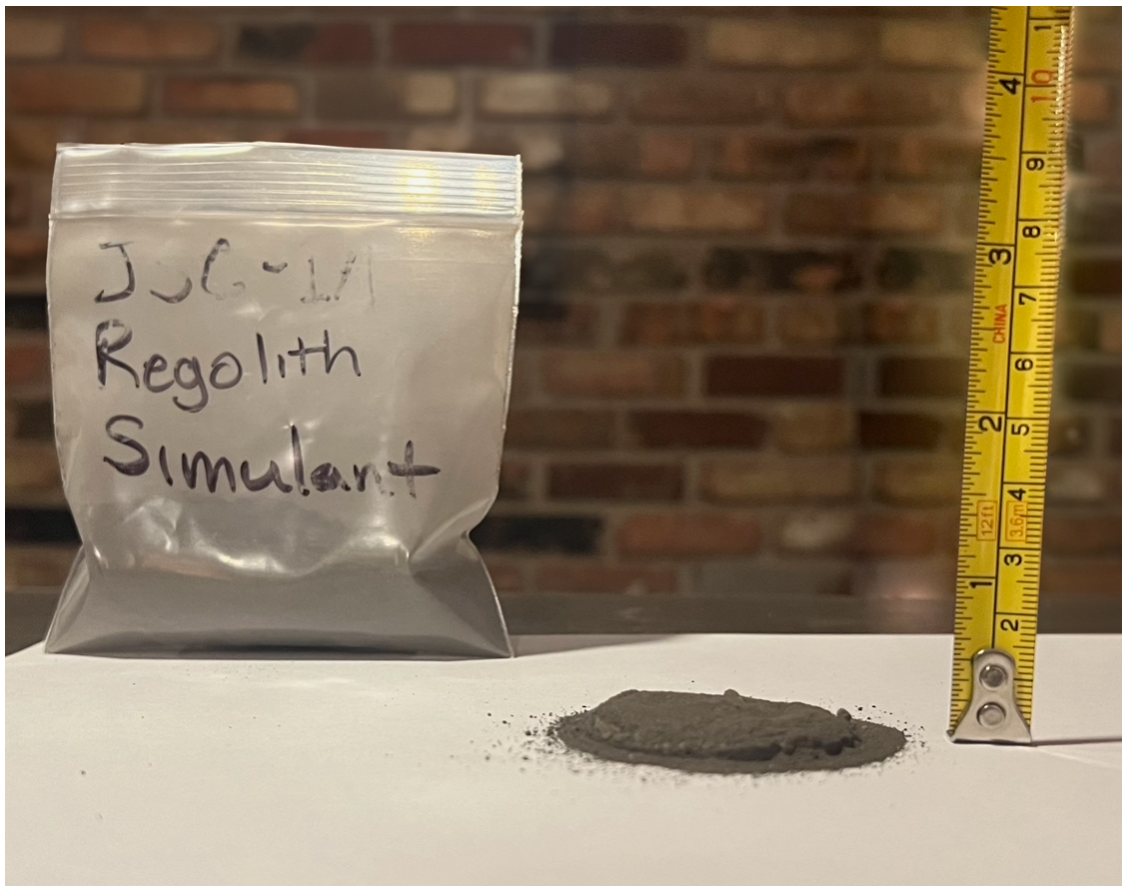


Fig. 9 Static angle of repose for physical JSC-1A experiment

When measuring the dimensions of the JSC-1A particle pile, the height is determined to be approximately 1 mm, while the length from the right side of the base is about 17 mm. Using the angle of repose was calculated to be approximately 6 degrees. This result demonstrates that the physical experiment closely aligns with the simulated results, further validating the accuracy of the modeling approach. Larger quantities of particles typically result in a more stable and consistent pile, providing a more accurate representation of the material's natural angle of repose. The cylinder's dimensions, including its diameter and height also play a crucial role in setting the initial conditions of the test. A

cylinder that is too small may restrict particle movement, preventing the formation of a natural pile, whereas an overly large cylinder might fail to adequately contain the material for accurate measurement. These factors could contribute to the slight variations observed between the simulated and physical angle of repose values.

IV. Sintered Regolith Flow

Initially, plans to utilize Lunar regolith for construction involved mixing the regolith particles with liquid binder materials to form a cohesive medium. While this approach seemed promising, it introduced several logistical and practical challenges. The primary issue was the need for a continuous supply of binder material, which would have to be transported from Earth, significantly increasing mission weight and costs. Furthermore, this method required considerable human intervention, both for precise mixing and application, making it less suitable for autonomous operations on the Moon. Adding to these challenges is the difficulty in identifying a suitable binder material. For instance, UV resin was considered due to its ability to rapidly cure and harden using the intense ultraviolet radiation from the sun [22]. However, it is ultimately ruled out due to its brittle nature when exposed to prolonged UV radiation, an issue that is inevitable given the Lunar surface's unabated subjection to solar radiation. These limitations highlighted the need for more self-sustaining solutions, such as direct sintering of Lunar regolith, which eliminates the dependency on additional materials and reduces the complexity of the process, aligning better with in-situ resource utilization principles for Lunar exploration.

The static angle of repose test confirms that the free-flowing particle behavior of the Lunar regolith simulant makes it an ideal candidate for consistent powder feeding—an essential factor in additive manufacturing systems. Building on this confirmation, the next step is to visually model how sintered Lunar regolith simulant flows out of a hopper to evaluate uniform distribution for stable layer formation. If the regolith doesn't flow smoothly under lunar gravity, clumping or interruptions occur, compromising the quality of sintered structures.

Sintering Lunar regolith is a feasible approach on the Moon, taking advantage of the natural vacuum conditions and the availability of concentrated solar energy through a solar furnace. By utilizing these existing Lunar resources, it becomes possible to create a new material, often referred to as "black glass," which can be shaped into durable building bricks. However, achieving this requires melting or sintering the regolith, a process that introduces new flow dynamics on the Moon. These dynamics are thoroughly studied to ensure the method is effective under Lunar conditions and can operate autonomously with minimal human intervention.

Figure 10 presents a flowchart outlining the configuration settings for modeling fluid flow. As with the physical particles, a new material type was created for the fluid, detailed in the left flowchart, where density data is inputted. Next, the smooth particle hydrodynamics (SPH) of the sintered regolith can be modeled, as illustrated in the right flowchart. SPH uses kernel approximation techniques to calculate densities and pressure gradients from a discrete set of freely moving particles [23]. The kernel approximation is obtained by substituting the Dirac delta distribution by an interpolation function such that:

$$\langle f(x) \rangle = \int_{\Omega} f(x') W(\|x - x'\|, h) d^n x', \quad (2)$$

Where $\langle f(x) \rangle$ represents the smoothed estimate of the function $f(x)$, $W = W(\|x - x'\|, h)$ is the interpolation or smoothing kernel, and h is the smoothing length. The kernel function W is typically a bell-shaped function that satisfies the following normalization condition of the SPH interpolation theory:

$$f(x) = \int_{\Omega} f(x') \delta(x - x') d^n x'$$

Where $f(x)$ is some smooth function defined in the domain $\Omega \subset \mathbb{R}^n$ and evaluated at the generic point $x \in \Omega$, $\delta(x - x')$ is the Dirac delta distribution, and n denotes the spatial dimension [24]. The kernel size of 0.001 m was selected based on the cumulative distribution chart, where this value best represented the particle size range relevant for the simulation. Additionally, a quintic kernel type was chosen because it closely approximates the behavior of molten glass, which is analogous to the properties of sintered regolith under the conditions being modeled [25].

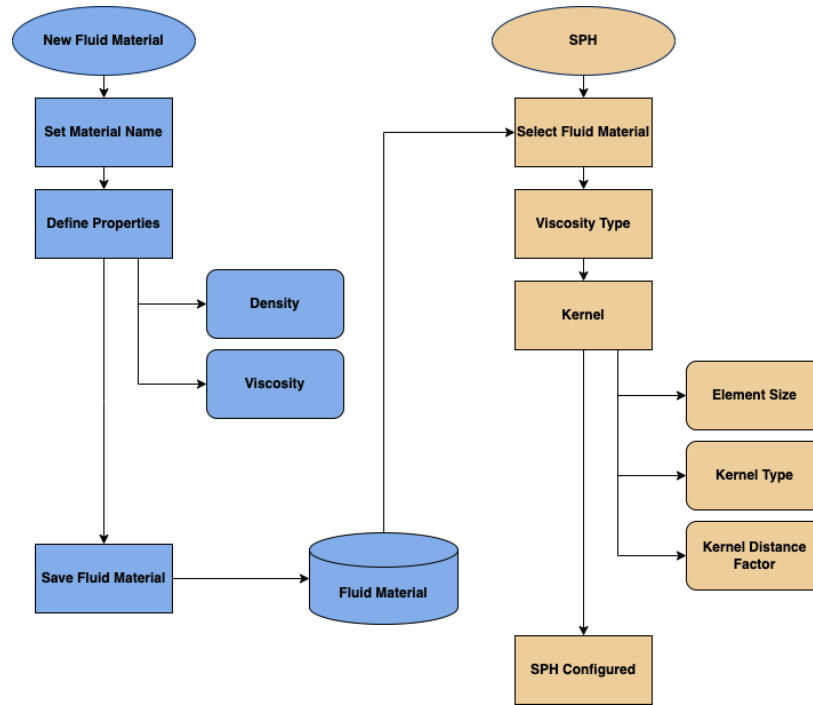


Fig. 10 ANSYS Rocky sintered fluid configuration flow

A. Lunar Vacuum

Light gases, like hydrogen, are heated to velocities sufficiently high enough to escape the gravitational pull of the Moon while other gases are eventually removed by the solar wind. As a result there is essentially no atmosphere to create an atmospheric pressure on the surface (at 14.69 psi), creating a hard vacuum [26]. The International Space Station on the contrary does not have these properties, resembling the Earth's atmosphere with 14.5 psi pressure and a composition of 79% nitrogen and 21% oxygen [27].

Using a vacuum environment significantly aids the solar sintering process for Lunar regolith due to the lower boiling point of minerals in a vacuum, which allows them to melt at lower energy thresholds compared to Earth's atmospheric conditions [28]. The challenges of testing Lunar regolith behavior on Earth involve limitations of available resources and equipment, such as vacuum chambers, which serve as the primary method for simulating Lunar conditions. Unlike the International Space Station (ISS), which has limitations in accessibility, cost, and the inability to house large-scale equipment, vacuum chambers on Earth allow for controlled experimentation under Lunar-like conditions. However, these chambers are not without flaws. Vacuum chambers require consistent maintenance to ensure reliable results. Various components must be inspected frequently for wear or cracks, as these can lead to seal failures. Additionally, vacuum pumps used in these setups are prone to malfunctions, such as oil contamination or insufficient pressure due to leaks, which can impact the accuracy of the experiments. Cleaning the chambers and replacing worn components are necessary but time-intensive processes that can introduce variability in results [29].

To overcome these challenges and expand on experimental capabilities, simulation tools like ANSYS Rocky offer an effective alternative. They enable precise modeling of particle flow dynamics under both Earth and Lunar gravity, simulating conditions that vacuum chambers might not fully replicate. This allows researchers to optimize configurations and understand regolith behavior before conducting physical tests. These simulations bridge the gap between theoretical studies and practical applications, offering a reliable and cost-efficient method to validate hypotheses without depending entirely on vacuum chambers.

B. Solar Furnace

Solar sintering of Lunar regolith demonstrates the ability to build a brick out of the sintered regolith, taking shape layer by layer as it passes through a concentrated solar beam [30]. Fresnel lenses or parabolic mirrors can concentrate sunlight to achieve the high temperatures necessary for sintering. These solar concentrators may be designed to track

the Sun's path, providing consistent energy input and minimizing fluctuations throughout the sintering process [31]. Because of the Lunar regolith's material make-up discussed in previous sections, when sintering the particle a black glass will be created, holding similar properties to glass here on Earth, with similar compressive strengths but also a lot higher tensile strength [32].

JSC-1A simulant serves as an ideal material for destructive testing, such as sintering, due to its availability and similarity to Lunar regolith in composition and behavior. However, the scarcity and high value of authentic Lunar regolith limit its use in experiments exploring diverse capabilities. By first simulating various conditions and achieving desirable results, JSC-1A can then be used to physically test these outcomes. This approach ensures that testing with the simulant is both efficient and effective, laying the groundwork for potential future experiments with authentic Lunar regolith. Once sintering is complete, assessing the flowability of the newly sintered Lunar regolith is crucial to determine the handling techniques required before the material solidifies fully in the vacuum environment. ANSYS Rocky can demonstrate the material's behavior right after sintering, allowing for the development of optimized methods to handle the regolith during construction or other applications.

C. Black Glass Flow: Earth's Gravity

To demonstrate the sintered regolith's flow-ability, a hopper with a radius of 45 mm and a height of 22 mm was imported as a wall geometry into Rocky. The radius was selected to ensure sufficient volume for containing the regolith, while the height was chosen to allow for an adequate fill of particles. The opening was calibrated to prevent clogging, yet remained small enough to ensure a steady flow of material. The sintered regolith was modeled with a density of 2850 kg/m³, a viscosity of 4 Pa-s, and a kernel size of 1 mm as a volumetric inlet, filling the hopper to the top before running the simulation.

Figure 11 illustrates the behavior of sintered particles in a hopper, observed after 5 seconds of free-flow under Earth's gravity. The particles exhibit a relatively uniform downward movement, filling the bottom of the hopper and escaping through the opening. This dispersion pattern aligns with what is expected under Earth's gravity, where the stronger gravitational force quickly pulls the particles downward, allowing them to align in a consistent, predictable manner.

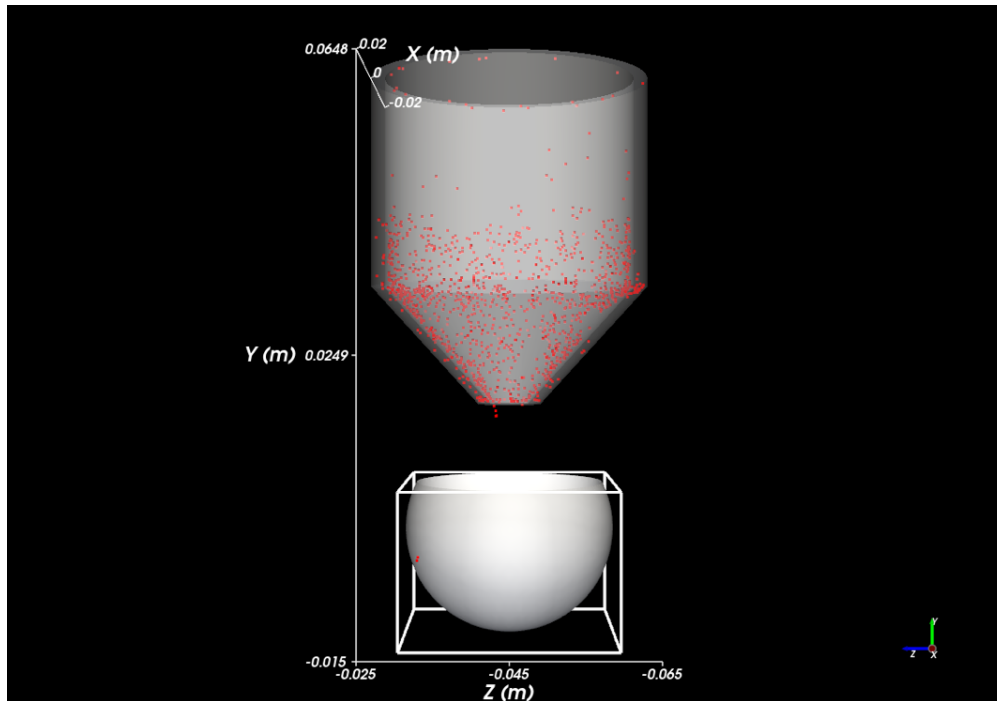


Fig. 11 Sintered regolith flow in Earth's gravity

The denser atmospheric pressure and gravitational force on Earth create a controlled environment for material flow dynamics. The cooling effect from Earth's atmosphere alters the viscosity of molten materials, contributing to more consistent flow behavior during additive manufacturing. A protective atmosphere is indispensable to a furnace because

the engagement of protective gas is beneficial to the formation of sintering necks between powders by reducing the content of oxides as residuals on particle surfaces [33]. This is less relevant in a vacuum where cooling is slower and thermal properties dominate flow behavior. As the particles flow, they exhibit a tendency to adhere to the walls of the hopper, forming an accumulation along the sides before finally falling. This behavior is a direct result of frictional forces between the particles and the hopper walls, which create resistance and slow the flow near the boundaries. These forces are crucial because they prevent the particles from moving as freely at the edges, leading to slight compaction and slower movement compared to the center of the flow. The flow pattern remains consistent across the entire cross-section of the hopper, without any noticeable clogging or disruptions. This uniformity is vital for applications like additive manufacturing, where consistent material deposition is necessary for the stable formation of layers. A uniform flow ensures that the material is distributed evenly, leading to a more reliable construction process.

D. Black Glass Flow: Lunar Gravity

To demonstrate in the Lunar gravity, the same hopper dimensions and sintered JSC-1A properties were used. The only variable changing was the gravity applied. Figure 12 illustrates the behavior of sintered particles in a hopper, observed after 5 seconds of free-flow under the Moon's gravity. Unlike Earth's conditions, where stronger gravity results in faster and more uniform flow, the reduced gravitational force on the Moon slows the movement of particles and alters their interaction with both each other and the hopper walls.

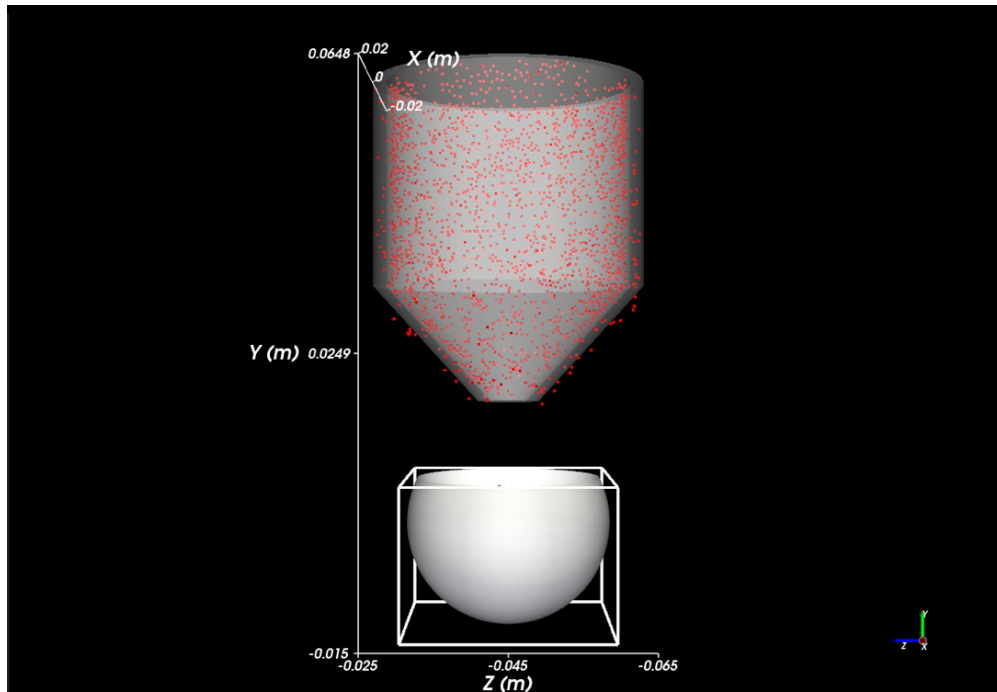


Fig. 12 Sintered regolith flow in Lunar gravity

In Lunar gravity, particles fall more slowly than on Earth, resulting in a more gradual flow within hoppers or during free dispersion. This reduced acceleration also decreases the inter-particle collisions and dispersive energy within the flow, leading to a smoother and more cohesive movement. Under Lunar gravity, particles exhibit stronger adhesion to container walls due to lower gravitational forces competing with frictional forces. This effect leads to significant boundary layer flow resistance, with particles often accumulating near the hopper walls [34]. The reduced Lunar gravity also slows the overall flow rate, leading to more irregular and scattered movement of particles through the nozzle. Unlike the consistent and fast flow observed under Earth's gravity, the weaker gravitational force creates a gentler flow pattern, with particles taking longer to move out of the hopper. This can result in less compact material deposition in the receiving area.

V. Conclusion

In the study of particle modeling and its applications, volcanic ash and Lunar regolith provide valuable analogs for understanding material behavior in extreme environments. Tests like the static angle of repose help characterize the flow and stability of these particles under varying conditions. Sintered regolith, created through processes like heating with a solar furnace, demonstrates potential for construction in both Lunar and Earth atmospheres. Simulations conducted in Lunar vacuum conditions highlight differences in material properties when exposed to the Moon's gravity compared to Earth's. By mapping and comparing simulated data from ANSYS Rocky with real-world results, this research refines techniques and guides future studies. This groundwork not only assesses the feasibility of using sintered regolith for construction but also supports formal requests to access Apollo mission Lunar regolith samples. These samples could facilitate advanced experiments, deepening our understanding of in-situ Lunar resource utilization and advancing strategies for sustainable Lunar exploration and habitation. The validated concept for Lunar regolith sintering using the JSC-1A simulant establishes a solid foundation for further physical experimentation. This research demonstrates the importance of precise modeling, including configurations for inputting particle geometries and defining specific particle distribution sizes. It also highlights the role of imported wall geometries in simulations, leveraging both Earth's gravity and Lunar gravity to ensure accuracy. Additionally, the study incorporates modeling the flow dynamics of sintered regolith particles, providing critical insights into material behavior during the sintering process.

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